

ABHIJAY GANGARAPU

Incoming Freshman | McNeil High School

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EDUCATION

McNeil High School

Austin, TX

Class of 2030 | Expected Graduation: May 2030

- Incoming Freshman (completed 8th grade)
- Two grades ahead in mathematics

RESEARCH INTERESTS

Number Theory • Algebra • p-adic Analysis • Algebraic Structures • Applied Mathematics • Computational Mathematics • Cryptography • Cybersecurity

PROJECTS

Algebraic Surjectivity and Super-Linear Period Growth in p-adic Piecewise Affine Maps

Independent Research | Number Theory & Neural Learning Limits

2025–2026

- Proved monodromy-based lift formula for period growth in piecewise affine maps over $\mathbb{Z}/p^k\mathbb{Z}$ with Hensel lifting
- Identified fundamental learning threshold $T/L_{\text{in}} \approx 35\text{--}150$ where gradient-based neural networks fail on long-period algebraic sequences
- Validated across 168 primes ($p = 2$ to $p = 997$) with mean growth ratio $26.94\times$ (maximum $79.42\times$)
- Discovered $G = \langle A_0, A_1 \rangle = \text{GL}_2(\mathbb{F}_p)$ for $p \in \{5, 7, 11, 13\}$ (full surjectivity) with 100% irreducibility verification
- Validated threshold on real ciphers: Trivium, ChaCha20, and LFSRs (6/6 correct predictions, 100% accuracy)
- Implemented in Python (NumPy, PyTorch, Numba JIT) with complete 140-page LaTeX paper and reproducible code
- **GitHub:** github.com/CoderAwesomeAbhi/neural-cryptanalysis

COMPETITIONS & AWARDS

Mathematics

- **AIME Qualifier** | American Invitational Mathematics Examination | 2026
- **AMC 10 Distinction** | Mathematical Association of America | 2026
- **MATHCOUNTS State Competitor** | Texas State Competition | 2026
- **Stanford Math Tournament (SMT) General:** Top 10% | 2026

Computer Science

- **USACO Silver Division** | USA Computing Olympiad | 2025–2026

Physics

- **USAPHO Qualifier** | USA Physics Olympiad | 2026

Science & Engineering

- **GARSEF 3rd Place Embedded Systems** | Engineering & General Sciences Division | 2026
- **GARSEF 4th Place** | Energy: Sustainable Materials and Design | 2024

TECHNICAL SKILLS

Programming: Python (intermediate), C++ (beginner), JavaScript

ML/AI: PyTorch, TensorFlow, scikit-learn, Keras

Scientific Computing: NumPy, SciPy, Matplotlib, Numba (JIT compilation)

Mathematical Tools: LaTeX, Jupyter, Mathematica

Development: Git, GitHub, VS Code, Docker

RELEVANT COURSEWORK & SELF-STUDY

Completed: Geometry (two grades ahead in math), Computer Science (Python, C++, algorithms)

Self-Study: p-adic Number Theory (Gouvêa's *p-adic Numbers: An Introduction*), Abstract Algebra, Number Theory (Hensel's Lemma, arithmetic dynamics, monodromy theory), Cryptography (pseudorandom generators, linear complexity, Berlekamp-Massey algorithm), Measure Theory (ergodicity, p-adic measures)

RESEARCH STATEMENT

I am passionate about applied mathematics, particularly number theory and algebraic structures. I'm eager to learn from experienced researchers and contribute to ongoing research projects.

What I Seek: Research mentorship where I can learn rigorous mathematical techniques and contribute to your research program. I'm open to working on any project you suggest—I want to learn how professional mathematical research is conducted.

What I Offer: Strong work ethic, solid programming skills (Python, C++), and genuine enthusiasm for mathematics and computer science. I'll do all coding, computations, and writing you need. I'm here to learn and contribute to your research goals.

Last Updated: May 13, 2026